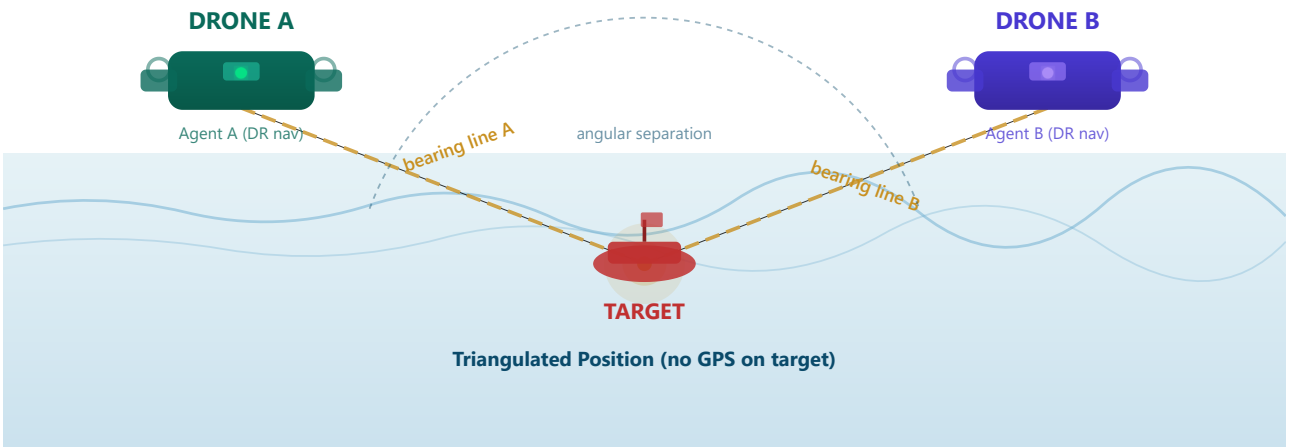
TECHNICAL PROPOSAL

GPS-Denied Multi-Agentic Sea Drone Triangulation

Autonomous Target Tracking via Bearing-Only Measurements, Dead Reckoning Navigation & Multi-Agent Coordination



Project: SeaDrone Agentic Simulation

Platform: NVIDIA Jetson + Pixhawk V6X + YOLOv8-nano + SAMURAI Tracker

Protocols: D2D (Drone-to-Drone) | D2G (Drone-Ground-Drone)

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Classification: RESTRICTED

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1. Executive Summary

This document defines the complete technical architecture for **GPS-denied multi-drone maritime target triangulation**. Two autonomous sea drones (Agent A, Agent B) collaboratively locate, track, and triangulate a target vessel using **only onboard sensors** — camera (YOLOv8-nano + SAMURAI tracker), IMU, compass — and inter-agent communication via a ground station orchestrator or direct mesh link.

> **Core Innovation:** A single drone can determine the *bearing* (direction) to a target but NOT the *range* (distance). Two drones observing from different positions produce two bearing lines whose **intersection resolves the target's position** — without any GPS on the target.

Key validated capabilities:

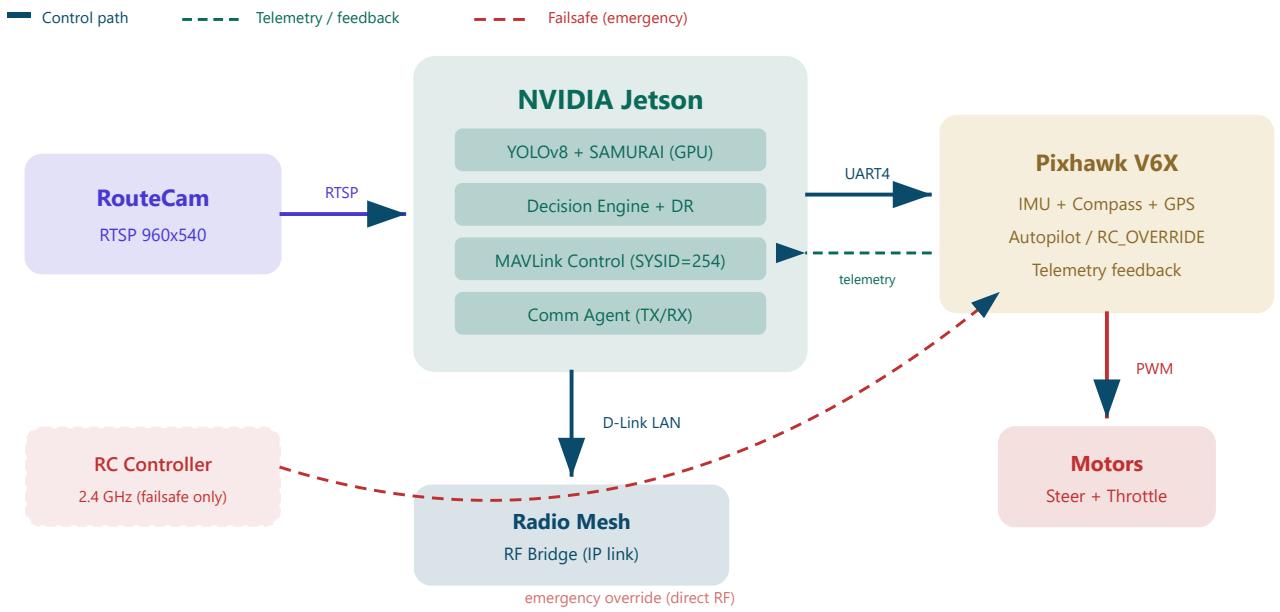
- Dead reckoning self-positioning: **3.4% drift over 800m** (field-validated)
- Camera-based target bearing: **directionally accurate** (YOLO/SAMURAI validated)
- Target loss recovery: **<30 seconds** via coordinated pincer search
- System latency: **<250ms** end-to-end (detection to drone response)
- Communication: **<64 bytes/sec** per drone (works on any radio link)

2. Hardware Platform & System Architecture

2.1 Per-Drone Hardware Stack

Component	Model	Role	Interface
Compute	NVIDIA Jetson (SeaDrone Lite)	AI inference, decision engine, MAVLink control	192.168.x.20, SYSID=254
Camera	RouteCam (RTSP)	Video stream to Jetson	192.168.x.10
Flight Controller	Pixhawk V6X	Autopilot, IMU, compass, GPS, motors	UART4 to Jetson, 192.168.x.41
Radio	Radio Mesh (RF Bridge)	IP bridge: drone ↔ GS and drone ↔ drone	192.168.x.x subnet
Network	D-Link Switch	Onboard LAN	192.168.x.x
Failsafe	RC Controller (2.4GHz)	Emergency override ONLY	Direct RF to Pixhawk

Per-Drone Hardware Block Diagram



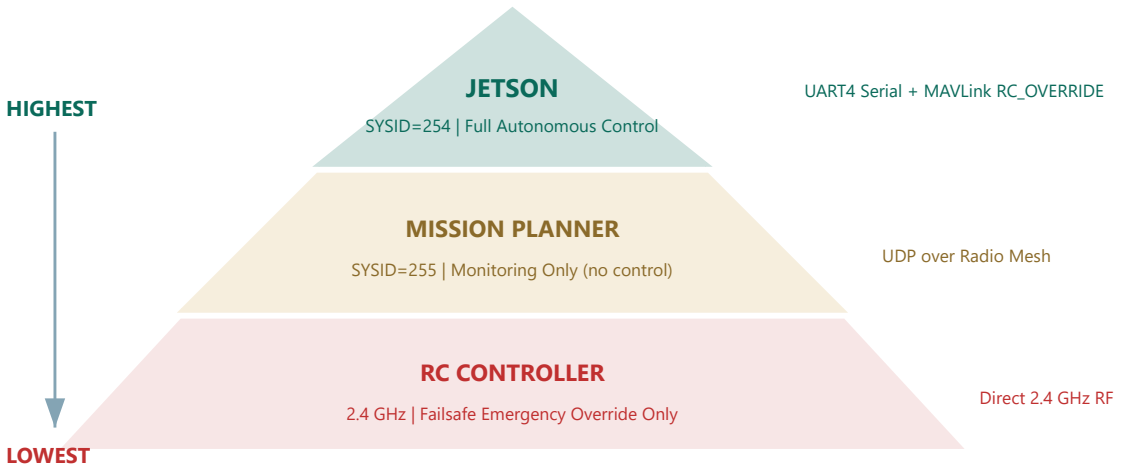
2.2 Ground Station

Component	Role	IP
Mac Workstation	GS Orchestrator Agent, monitoring, operator UI	192.168.1.160
Mission Planner	Telemetry monitoring (NO CONTROL authority)	SYSID=255
Radio Mesh A	Ground-side RF bridge	Via D-Link Switch

2.3 Control Authority Model

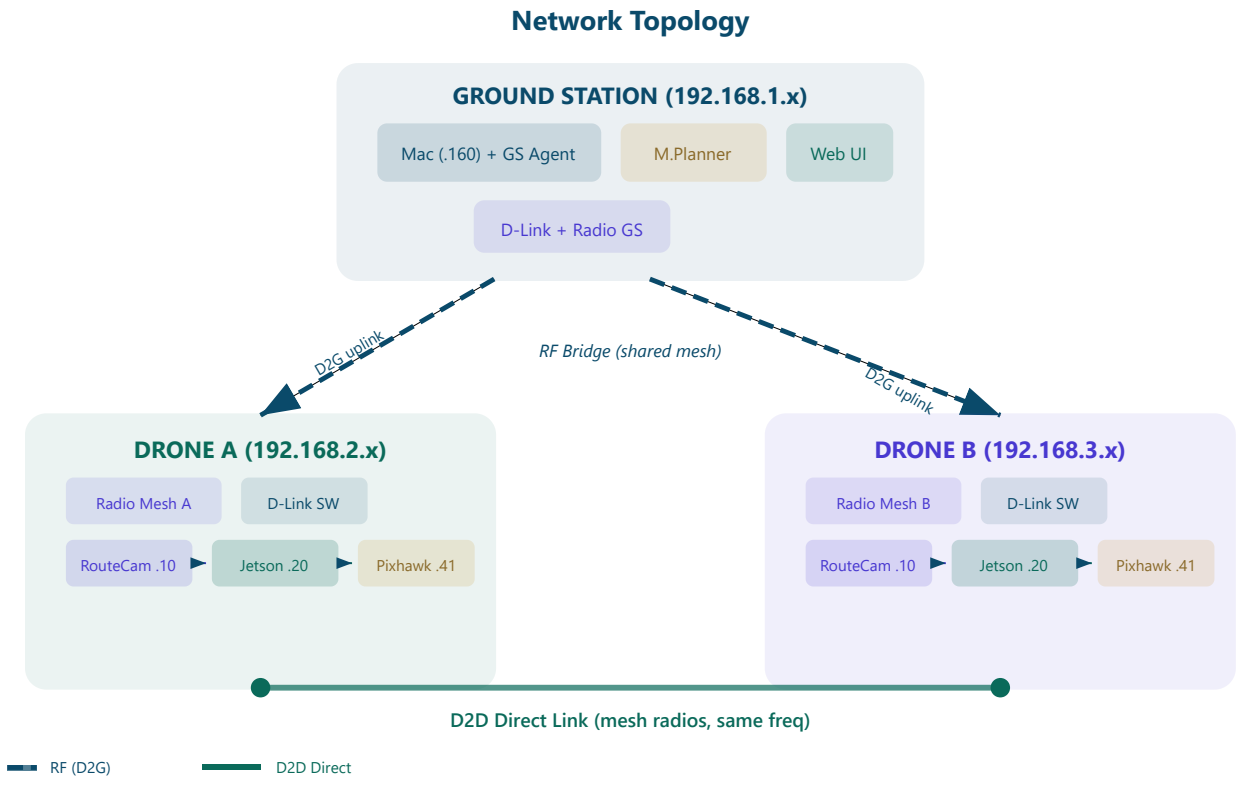
```
CONTROL PRIORITY (highest to lowest):

  1. Jetson (SYSID=254) — FULL AUTONOMOUS CONTROL
     via UART4 Serial → Pixhawk → MAVLink RC_OVERRIDE
     Commands: steering (ch1) + throttle (ch3) PWM values

  2. Mission Planner (SYSID=255) — MONITORING ONLY
     via UDP over Radio Mesh → No control authority

  3. RC Controller — FAILSAFE ONLY
     via 2.4GHz RF direct to Pixhawk
     Activates ONLY if Jetson link lost
```

Control Authority Priority Pyramid



2.4 Network Topology — Multi-Drone Extension

```
GROUND STATION (192.168.1.x)
┌──────────────────────────────────┐
│  Mac 192.168.1.160               │
│  ├─ GS Orchestrator Agent        │
│  ├─ Mission Planner (monitor)    │
│  └─ Web Dashboard                │
│       │                          │
│  D-Link Switch → Radio Mesh GS   │
└───────┼──────────────────────────┘
        │ RF Bridge (shared mesh)
    ┌───┴──────────────────┐
    │                      │
DRONE A (192.168.2.x)   DRONE B (192.168.3.x)
┌─────────────────┐      ┌─────────────────┐
│ Radio Mesh A    │      │ Radio Mesh B    │
│ D-Link Switch   │      │ D-Link Switch   │
│ RouteCam  Jetson │      │ RouteCam  Jetson │
│    .10     .20  │      │    .10     .20  │
│        Pixhawk  │      │        Pixhawk  │
│         .41     │      │         .41     │
└─────────────────┘      └─────────────────┘
          │                      │
          └──── D2D Direct Link ──┘
           (mesh radios, same freq)
```



Subnet separation: Each drone gets its own subnet to prevent IP conflicts. The mesh radios handle

> routing between subnets. D2D communication uses direct mesh radio without routing through GS.

3. Sensor Data Analysis & Field Validation

3.1 Platform Capabilities (Measured from Sea Trial)

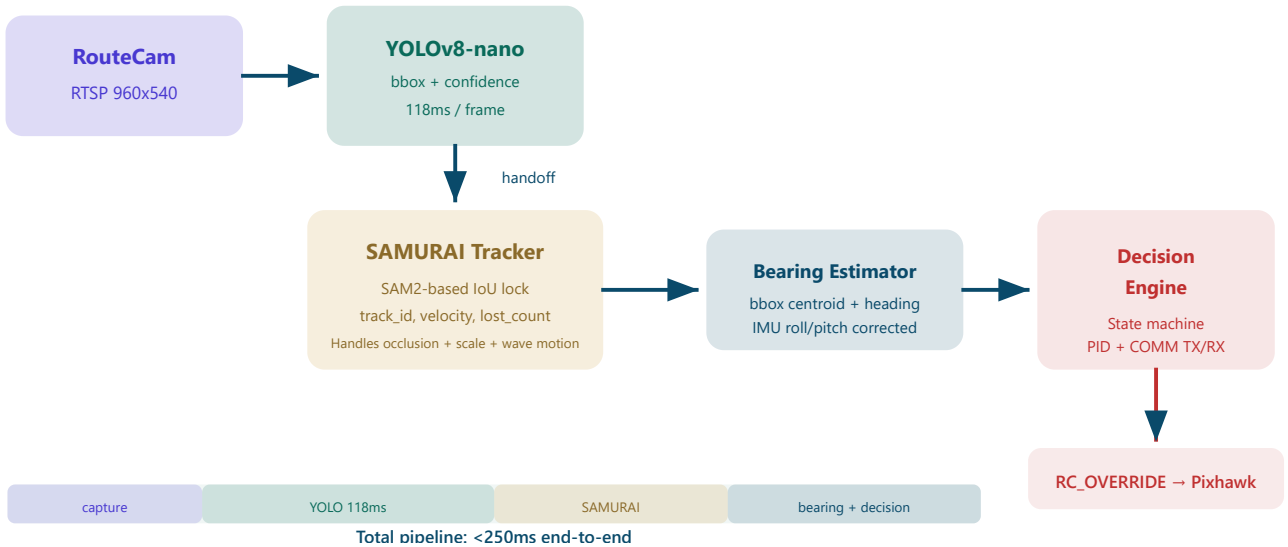
Based on `sea_drone_data_v1.json` — 394 frames from a real sea drone tracking session:

Parameter	Value	Source
Frame rate	~1.3 FPS	394 frames / ~300 seconds
Detection model	YOLOv8-nano → SAMURAI	Hybrid detection + tracking
Inference time	~118ms per frame	Jetson GPU
Camera resolution	960 × 540	RouteCam RTSP
Ground speed	0.6 — 3.6 m/s	Pixhawk VFR_HUD
Detection confidence	0.45 — 0.81	YOLOv8 output
Tracking velocity	-274 to +90 px/s (X)	SAMURAI tracker
Detection rate	269/394 frames (68%)	With lock/lost transitions
Path distance	797m (128m straight-line)	GPS ground truth
Battery consumption	3% per 5 minutes	Pixhawk SYS_STATUS

3.2 Detection & Tracking Pipeline

```
RouteCam (RTSP, 960×540)
   │
   ▼
┌─────────────────┐
│  YOLOv8-nano    │  Initial detection: bbox + confidence + "boat" class
└─────────────────┘
        │ handoff to tracker
        ▼
┌─────────────────┐
│  SAMURAI        │  SAM2-based persistent tracking
│  (IoU Lock)     │  Maintains track ID across:
│                 │  - Partial occlusion
│                 │  - Scale changes (distance)
│                 │  - Camera motion from waves
│                 │  Outputs: track_id, velocity_px/s, lost_count
└─────────────────┘
        │
        ▼
  Bounding Box → Bearing Estimation → Decision Engine
```

Detection & Tracking Pipeline



> **Operator interaction:** The operator performs "Click to Lock" for initial target designation. After that, SAMURAI maintains autonomous tracking. The operator can also Arm/Launch/Stop the mission.

3.3 Dead Reckoning Feasibility (Field Validated)

Test	Method	Result
Self-Position (DR vs GPS)	Compass heading + ground speed integration	**3.4% drift over 800m in 5 minutes**

Target Bearing	YOLO/SAMURAI bbox centroid + drone heading	**Directionally accurate**
Target Range	Bbox size → range estimate	**Approximate (±30-50%)** — needs triangulation

> **Key finding:** A single drone can determine target *direction* but NOT target *distance*. This is the fundamental reason two-drone triangulation is required — two bearings from different positions geometrically resolve the range.

DR Error Model for Triangulation

```
DR position error after traveling distance D:
  σ_position = 0.034 × D

Example: drone traveled 200m since last known fix
  σ_position = 6.8m
  At 50m target range → ~7.7° bearing line origin error
  At 100m target range → ~3.9° bearing line origin error

Mitigation: weight each drone's bearing by inverse of DR age
```

4. The GPS-Denied Problem

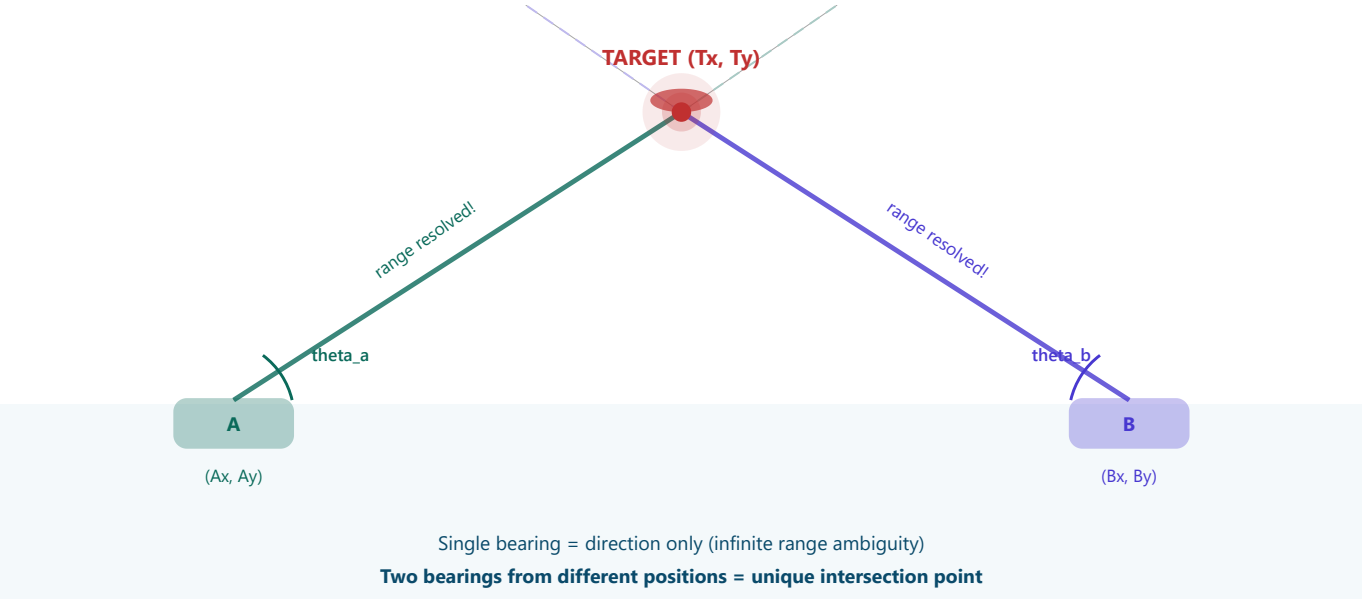
4.1 Known vs Unknown

Known (per drone)	Unknown
Own position (dead reckoning from launch)	Target position (world coordinates)
Own heading (compass + IMU yaw)	Target heading
Own speed (ground speed sensor)	Target speed
Target bearing in camera (bbox centroid)	Target range/distance
Target pixel velocity (px/s)	Target world velocity
Detection confidence	Target identity confirmation

4.2 Why Triangulation Solves the Range Ambiguity

```
Single Drone:
  ✓ Own position (DR, 3.4% drift)
  ✓ Target BEARING (camera) — accurate direction
  ✗ Target RANGE — unknown

Two Drones:
  ✓ Both own positions (each via DR)
  ✓ Two independent BEARINGS from different positions
  ✓ Bearing intersection → TARGET POSITION resolved
  ✓ RANGE determined geometrically — no calibration needed
```

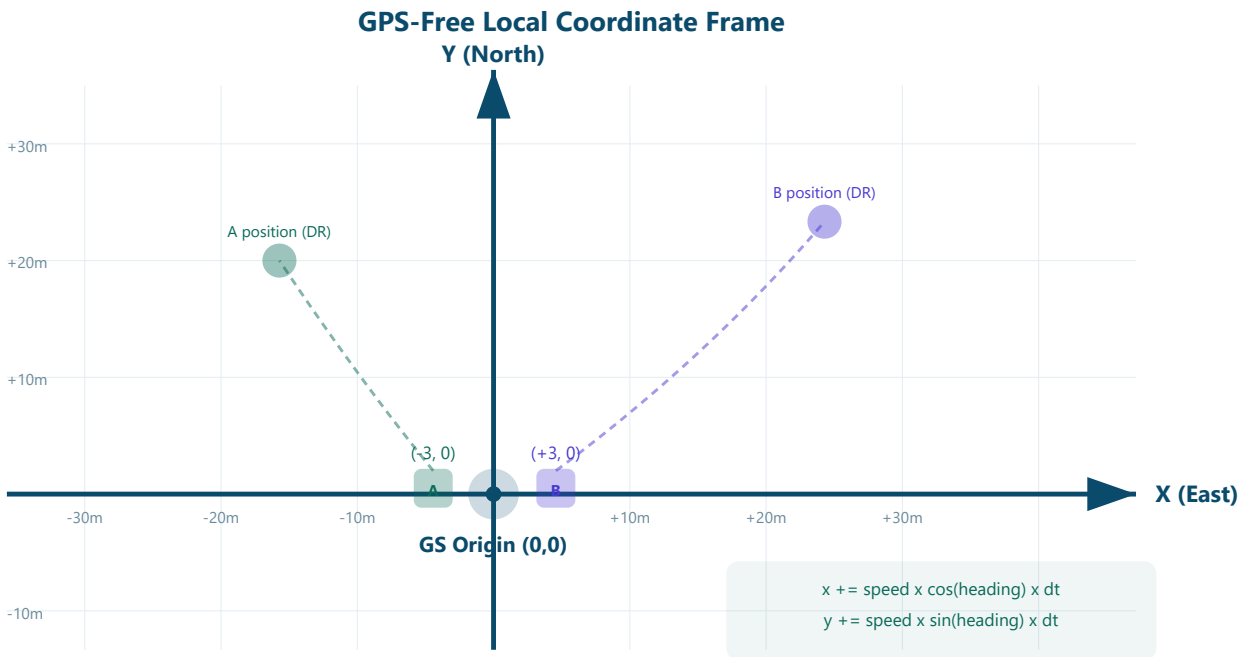
Triangulation: Two Bearings Resolve Target Position



5. Coordinate System — GPS-Free Local Frame

All positions expressed in **meters East/North from the ground station (launch point)**. No latitude/longitude anywhere in the protocol.

```
Origin:  Ground Station position = (0, 0)
X-axis:  East (positive = right)
Y-axis:  North (positive = up)

Drone A launch: (-3, 0) meters from GS
Drone B launch: (+3, 0) meters from GS

Position update (each timestep dt):
  x += speed × cos(heading) × dt
  y += speed × sin(heading) × dt

Position quality tag (sent with every message):
  dr_distance_m:     total distance traveled on DR
  estimated_error_m: dr_distance × 0.034
  age_sec:           time since last absolute calibration
```

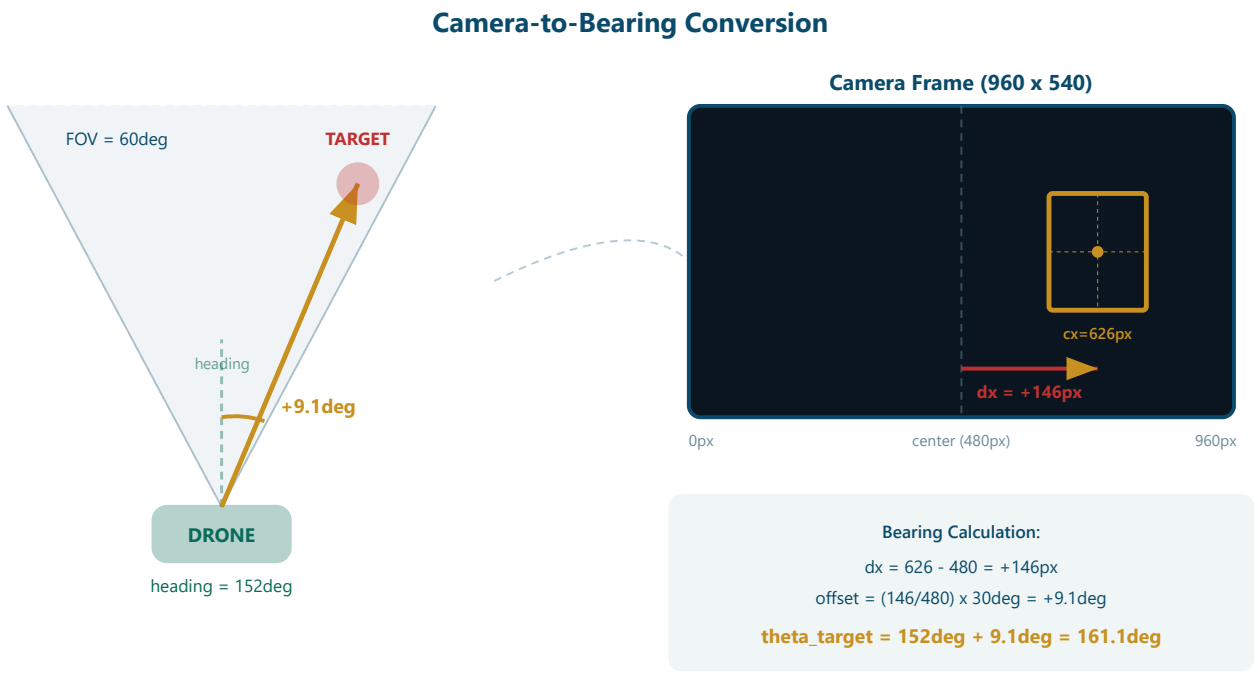


All positions in meters from GS. No lat/lon in protocol.

6. Bearing Estimation from Camera Data

6.1 Camera-to-Bearing Conversion

```
Target bbox center: (cx, cy) in pixels
Camera FOV horizontal: FOV_H = 60°
Frame width: W = 960 pixels

Pixel offset from center: dx = cx - (W / 2)
Bearing offset: θ_offset = (dx / (W/2)) × (FOV_H / 2)
Absolute bearing: θ_target = drone_heading + θ_offset

Example from real data:
  bbox center x = 626, heading = 152°
  dx = 626 - 480 = +146px (target right of center)
  θ_offset = (146/480) × 30° = +9.1°
  θ_target = 152° + 9.1° = 161.1°
```



6.2 Bearing Confidence & Wave Compensation

```
corrected_bearing = raw_bearing - roll_component × sin(pitch)
bearing_confidence = detection_confidence × (1 - |roll| / 30°)

From real data: roll up to 5.6°, pitch up to 8.6°
→ Systematic bearing error correctable with IMU data
```

6.3 Bearing Rate for Target Motion Prediction

```
bearing_rate = (velocity_px_per_sec.x / (W/2)) × (FOV_H / 2)   deg/sec

Real data: velocity_px_per_sec.x ranges -274 to +90 px/s
→ bearing_rate ranges roughly -17 to +5.6 deg/sec

This tells us:
  Positive rate → target moving right in camera → moving left relative to drone heading
  Negative rate → target moving left in camera → moving right relative to drone heading
  High |rate| → target crossing fast → LEAD INTERCEPT required
```

7. Communication Protocol

7.1 Five Message Types (Compact Binary, GPS-Free)

Message 1: BEARING_REPORT (32 bytes, 1 Hz when tracking)

The single most important message for triangulation.

Field	Type	Bytes	Description
msg_type	uint8	1	= 0x01
sequence	uint16	2	Rolling counter
source_agent	uint8	1	0=A, 1=B
timestamp_ms	uint32	4	ms since mission start
source_x_m	float32	4	Own position, meters east from GS
source_y_m	float32	4	Own position, meters north from GS
source_heading_deg	float16	2	Compass heading
source_speed_mps	float16	2	Ground speed
target_bearing_abs_deg	float16	2	**Absolute bearing to target**
target_bearing_rate_dps	float16	2	Bearing rate (deg/sec)
detection_confidence	uint8	1	0-100 scaled
bbox_area_px	uint16	2	For range estimation
bbox_cx_px	uint16	2	Bbox center x in frame
dr_quality	uint8	1	0-100 (100=fresh, 0=max drift)
track_lost_count	uint8	1	0=solid lock
padding	uint8	1	

Total: 32 bytes — at 9600 bps radio: 27ms transmission time. Supports 1 Hz update on any practical link.

Message 2: TARGET_LOST (24 bytes, event-triggered)

Field	Type	Bytes	Description

msg_type	uint8	1	= 0x02
source_agent	uint8	1	
timestamp_ms	uint32	4	
last_bearing_deg	float16	2	Bearing at moment of loss
bearing_rate_at_loss	float16	2	**Predicts escape direction**
estimated_range_m	float16	2	From bbox size heuristic
loss_reason	uint8	1	0=SIDE, 1=AHEAD, 2=OCCLUDED, 3=EVADED
source_x_m	float32	4	
source_y_m	float32	4	
source_heading_deg	float16	2	
padding	uint8	1	

Message 3: INTERCEPT_COMMAND (28 bytes)

Field	Type	Bytes	Description
msg_type	uint8	1	= 0x03
target_agent	uint8	1	
timestamp_ms	uint32	4	
waypoint_x_m	float32	4	Where to go
waypoint_y_m	float32	4	
approach_heading_deg	float16	2	Heading to arrive with (FOV aimed at target)
approach_speed_mps	float16	2	
predicted_target_x_m	float32	4	Where target will be
predicted_target_y_m	float32	4	
urgency	uint8	1	0=LOW, 1=MED, 2=HIGH, 3=CRITICAL
padding	uint8	1	

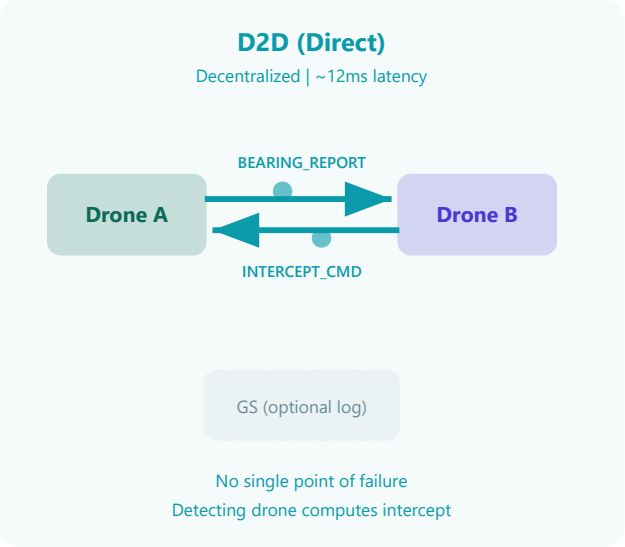
Message 4: TRIANGULATION_FIX (20 bytes)

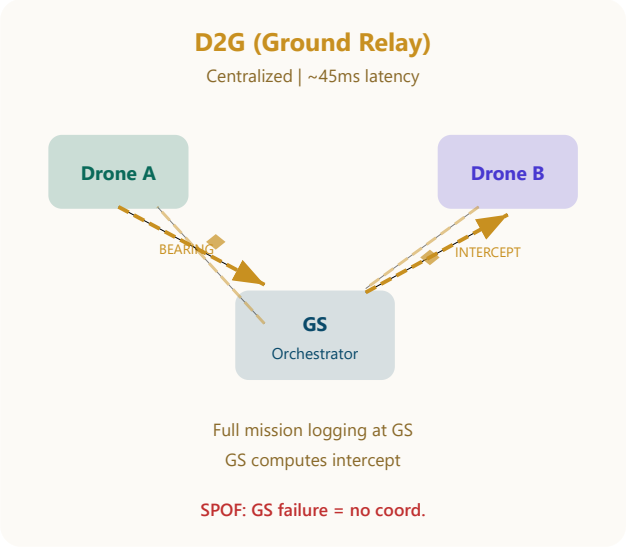
Message 5: SEARCH_ASSIGN (20 bytes)

See Appendix A for full field specifications.

7.2 D2D vs D2G Protocol Modes

Aspect	D2D (Direct)	D2G (Ground Relay)
Routing	A ↔ B direct	A → GS → B
Latency	~12ms	~45ms
Bandwidth	0.8 Mbps	2.4 Mbps
Coordinator	Detecting drone computes intercept	GS computes intercept
Single point of failure	None — decentralized	GS failure = no coordination
Data logging	No central log	Full mission logging at GS
Range	Limited by D2D radio range	Full operational area
Visual (simulation)	Cyan solid beam, sphere packets	Amber dashed beams via GS, diamond packets



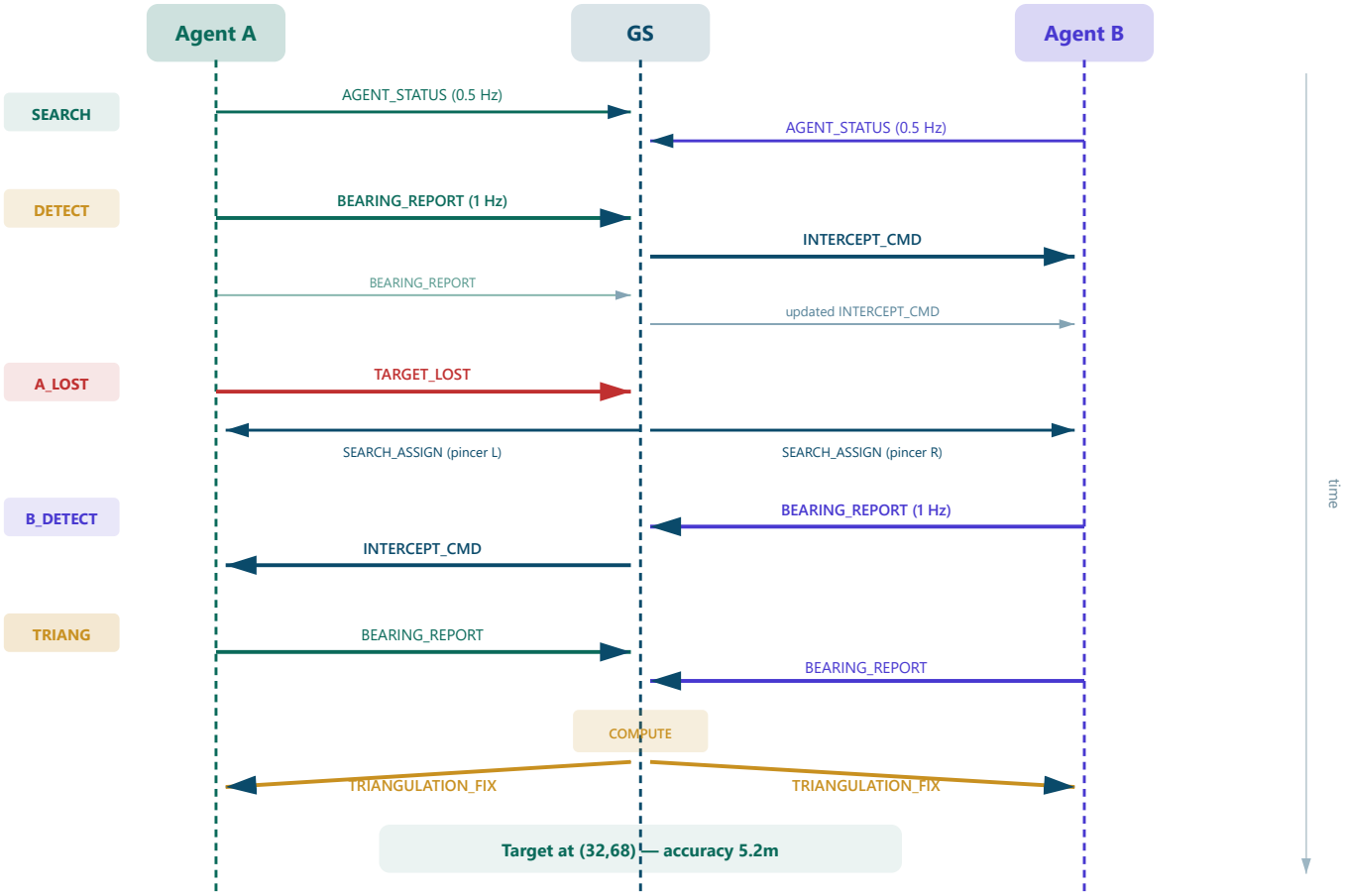


> **User-selectable:** The protocol mode is chosen by the operator before or during mission via the dashboard selector. This allows demonstrating both approaches to stakeholders.

7.3 Communication Flow by Mission Phase

```
SEARCH:     A,B → GS: AGENT_STATUS (0.5 Hz)    "searching, no contact"
DETECT:     A → GS:   BEARING_REPORT (1 Hz)     "contact, bearing 161°, rate -2°/s"
            GS → B:   INTERCEPT_CMD              "go to (45,80), approach 245°"
A_LOST:     A → GS:   TARGET_LOST               "lost, last bearing 175°, rate -5°"
            GS → A,B: SEARCH_ASSIGN              "pincer search zone (30,65) r=20m"
B_DETECT:   B → GS:   BEARING_REPORT (1 Hz)     "contact, bearing 72°, conf 0.71"
            GS → A:   INTERCEPT_CMD              "navigate to (35,70), approach 320°"
TRIANG:     A,B → GS: BEARING_REPORT (1 Hz)     dual bearings
            GS → A,B: TRIANGULATION_FIX          "target at (32,68), accuracy 5.2m"
```

Communication Flow — Sequence Diagram



8. Triangulation Algorithm

8.1 Bearing-Only Intersection

```
Agent A at (Ax, Ay) with bearing θa
Agent B at (Bx, By) with bearing θb

Bearing line A: (x,y) = (Ax,Ay) + t × (cos θa, sin θa)
Bearing line B: (x,y) = (Bx,By) + s × (cos θb, sin θb)

Intersection parameter:
  t = ((Bx-Ax)×sin θb - (By-Ay)×cos θb) / (cos θa×sin θb - sin θa×cos θb)

Target position:
  Tx = Ax + t × cos θa
  Ty = Ay + t × sin θa
```

Bearing-Only Triangulation & GDOP

8.2 GDOP & Accuracy

Angular Separation	GDOP	Quality
180°	1.00	OPTIMAL
120°	1.15	EXCELLENT
90°	1.41	GOOD
60°	2.00	FAIR
30°	3.86	POOR

Minimum viable separation: 60°. Optimal: 90°–180°.

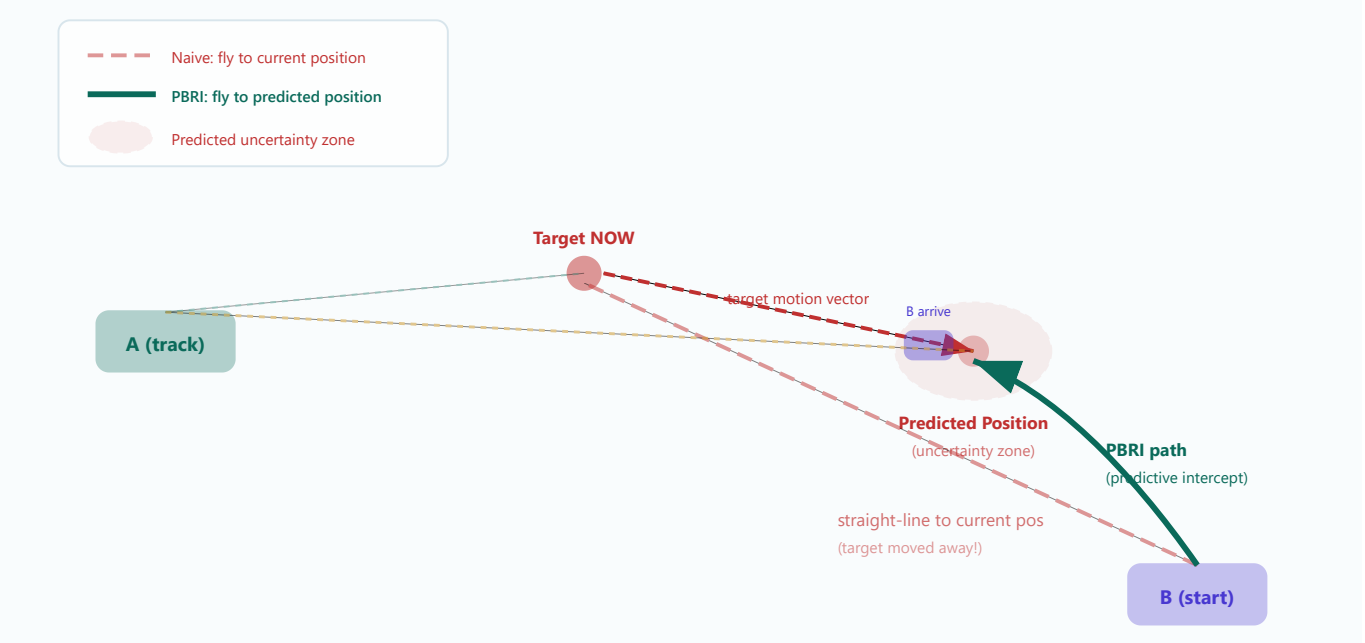
9. Agentic Path Planning

9.1 Predictive Bearing-Rate Intercept (PBRI)

> **Key principle:** Don't fly to where the target IS — fly to where it WILL BE. Use bearing rate history to predict target motion and compute an optimal intercept point that maximizes triangulation geometry.

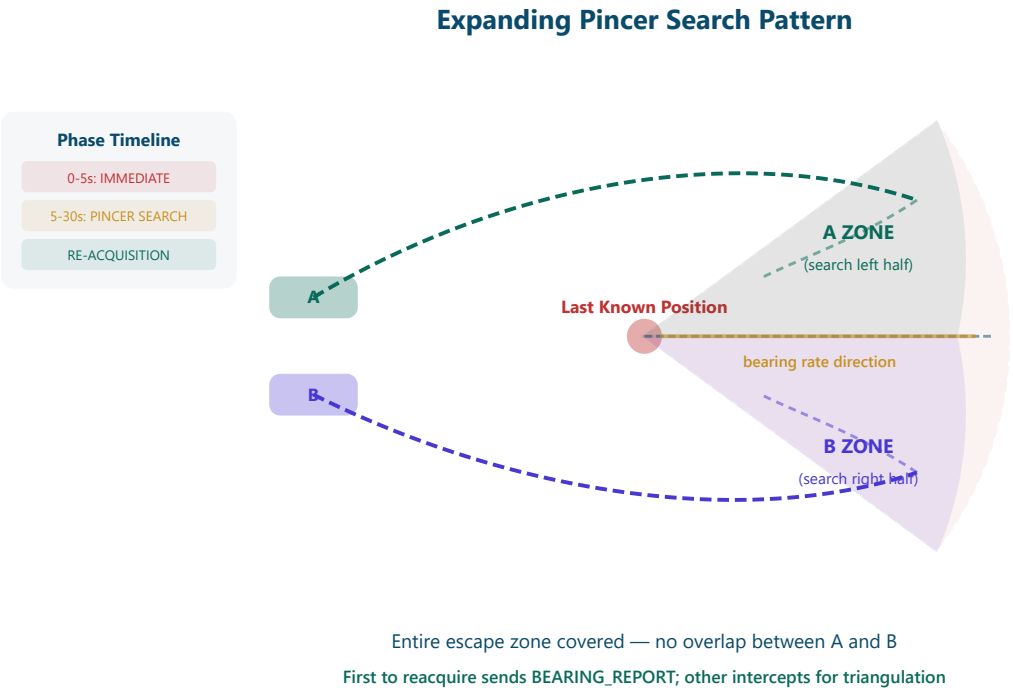
```
PBRI Algorithm:

Step 1: ESTIMATE TARGET MOTION
  Input: N bearing reports from Agent A over time window T
  Compute: bearing rate ω = dθ/dt (linear regression)
  Classify: |ω| < 0.5°/s → STATIONARY
            |ω| 0.5-3°/s → SLOW
            |ω| 3-8°/s   → MEDIUM
            |ω| > 8°/s   → FAST / TURNING

Step 2: COMPUTE PREDICTED TARGET ZONE
  Target bearing at T_future: θ_future = θ_now + ω × ΔT
  Range estimate: R = K / √(bbox_area)
  Predicted position: P = P_A + R × (cos θ_future, sin θ_future)
  Uncertainty: ±5° + 2°/sec × time_since_detection

Step 3: SCORE INTERCEPT CANDIDATES
  For each candidate arrival time T:
    score = w1 × triangulation_quality(angular_sep)
          + w2 × arrival_margin(time_slack)
          + w3 × fov_coverage(zone_size vs FOV)
          - w4 × energy_cost(distance / battery)

Step 4: GENERATE CURVED PATH
  Dubins path from B's current position to intercept point
  Arrive facing the predicted target zone (FOV pre-aimed)
```

PBRI: Predictive vs Straight-Line Intercept



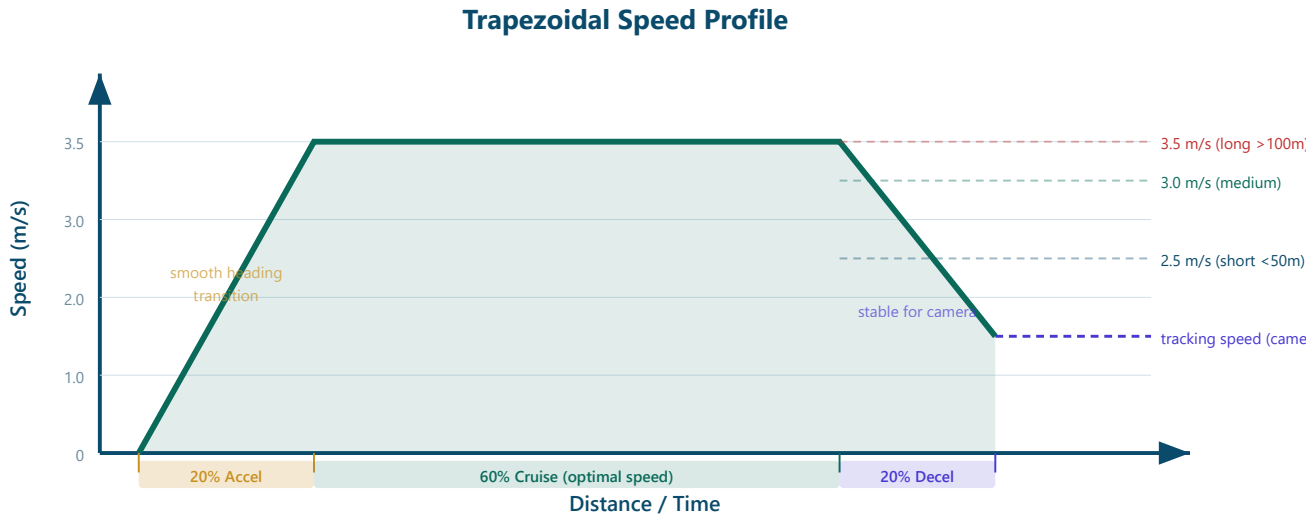
9.2 Target Loss Recovery — Expanding Pincer Search

```
Phase 1: IMMEDIATE (0-5 sec)
  A: Turn toward last bearing rate direction
  A: Broadcast TARGET_LOST with escape prediction
  B: Receive and compute predicted escape zone

Phase 2: COORDINATED SEARCH (5-30 sec)
  Escape zone = wedge from last known position
  A: Search LEFT half of escape wedge
  B: Search RIGHT half of escape wedge
  → Entire probable zone covered, no overlap

Phase 3: RE-ACQUISITION
  Whichever drone finds target → BEARING_REPORT
  Other drone → intercept for triangulation
```



9.3 Energy-Optimal Speed Profile

```
Trapezoidal profile:
  Phase 1 (20%): Accelerate — smooth heading transition
  Phase 2 (60%): Cruise at optimal speed
  Phase 3 (20%): Decelerate to tracking speed (stable camera)

Speed selection:
  Short (<50m):   2.5 m/s (accuracy priority)
  Medium (50-100m): 3.0 m/s (balanced)
  Long (>100m):    3.5 m/s (speed priority)
  Critical:        3.6 m/s max
```

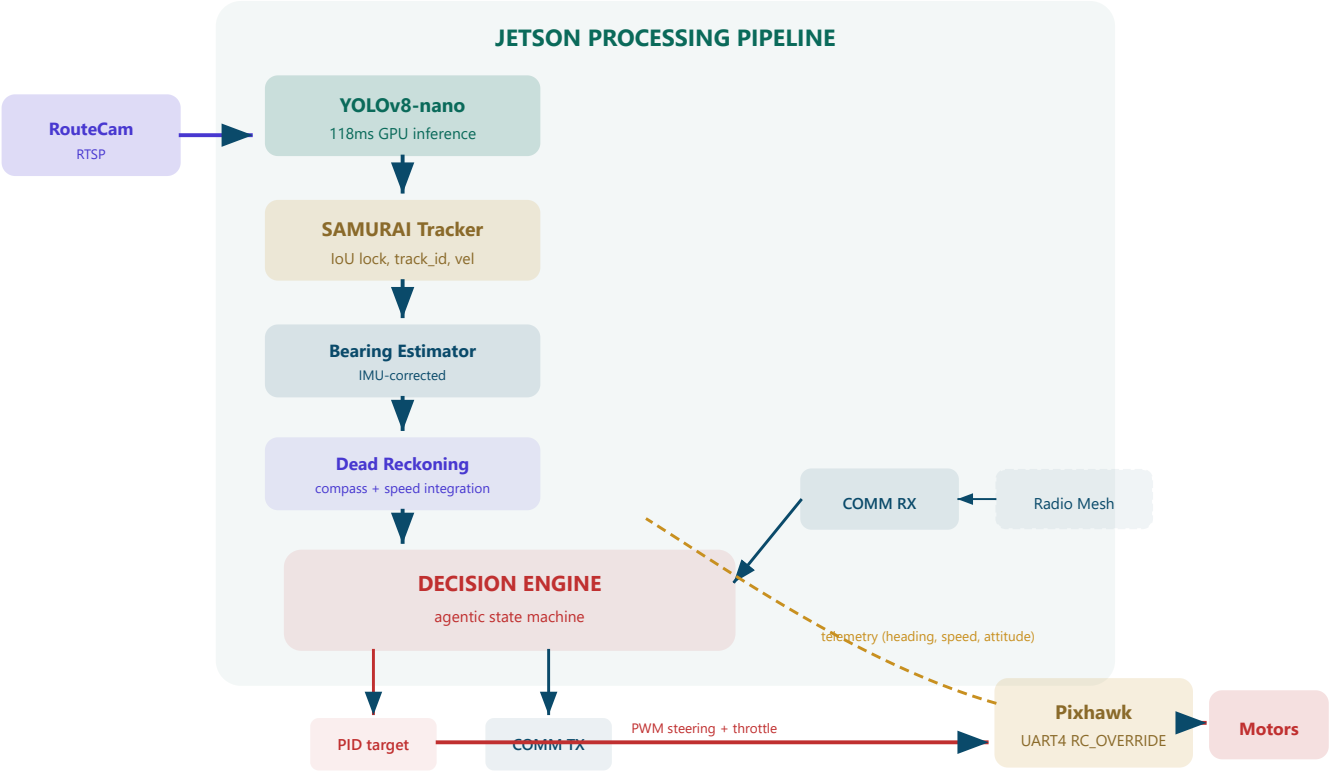


9.4 Arrival Heading Optimization

10. Onboard Processing Pipeline

```
RouteCam (RTSP) → Jetson GPU → Pixhawk (UART4) → Motors
                ┌──────────────────────────────┐
                │ JETSON PROCESSING PIPELINE    │
                │                              │
                │  YOLOv8-nano (118ms)          │
                │       ↓                       │
                │  SAMURAI Tracker              │
                │       ↓                       │
                │  Bearing Estimator (IMU-corrected)│
                │       ↓                       │
                │  Dead Reckoning (compass+speed) │
                │       ↓                       │
                │  ┌──────────────────┐         │
                │  │  DECISION ENGINE     │        │
                │  │  (agentic state      │        │
                │  │   machine)           │        │
                │  └──┬────────┬────────┘         │
                │     │        │        │         │
                │   PID     COMM    COMM          │
                │  target    TX      RX           │
                │     ↓                           │
                │  RC_OVERRIDE → UART4 → Pixhawk  │
                └──────────────────────────────┘
```

Onboard Processing Pipeline



> **Current → Proposed transition:** The existing PID controller is preserved. The Decision Engine wraps it — instead of always chasing the visual target, it can command the PID to follow search patterns, navigate to intercept waypoints, or orbit at standoff distance.

11. Mission Phase Information Flow

Phase	Agent A	Agent B	GS / Coordinator
DEPLOY	Launch, begin DR	Launch, begin DR	Log, assign search sectors
SEARCH	Sweep NW, AGENT_STATUS	Sweep NE, AGENT_STATUS	Monitor positions
A DETECT	Lock target, BEARING_REPORT	Searching (no contact)	Compute intercept → INTERCEPT_CMD to B
A RELAY	Track + send bearings	Navigate to intercept	Update intercept from new bearings
A LOST	Spiral search, TARGET_LOST	Continue toward area	SEARCH_ASSIGN: pincer L/R
B DETECT	Navigate to B's area	Lock target, BEARING_REPORT	INTERCEPT_CMD to A
TRIANGULATE	BEARING_REPORT (1 Hz)	BEARING_REPORT (1 Hz)	Compute intersection → TRIANGULATION_FIX
TRACK	Orbit 180° from B	Orbit 180° from A	Monitor GDOP, adjust orbits

12. Handling Real-World Challenges

Challenge	Impact	Mitigation
Latency	Target moves 0.36-0.43m during pipeline	Timestamp every message; GS extrapolates bearings to common reference time
Wave-induced bearing noise	Roll 5.6°, pitch 8.6° tilt camera	IMU-based correction; Kalman filter smoothing
Intermittent detection	32% miss rate	SAMURAI persistent tracking; bearing extrapolation during gaps; lost_count threshold
DR drift	3.4% over 800m	Weight bearings by DR freshness; periodic GPS recalibration when available
Battery	3% per 5 min → ~112 min endurance	Include in intercept planning — don't send low-battery drone on long intercept
Target evasion	Bearing rate changes sign	Loss classification → pincer search → re-acquisition
Radio interference	Packet loss	Sequence numbers for ordering; extrapolate from last known; 32-byte packets = fast retransmit

13. Performance Targets

Metric	Target	Basis
Detection to B intercept	< 60 seconds	B covers 180m at 3 m/s
Triangulation accuracy	< 10m CEP	GDOP 1.41 at 90° sep, 50m range, 5° bearing error
Target loss recovery	< 30 seconds	Pincer search covers 40m radius
Communication bandwidth	< 64 bytes/sec/drone	32-byte packet at 1 Hz + overhead
DR position accuracy	< 10m after 5 min	3.4% drift validated
Bearing accuracy	< 5°	960px frame, 60° FOV, SAMURAI centroid
System latency	< 250ms	Pipeline: 147ms onboard + comm + compute

14. Field-Validated Assumptions

Assumption	Validation	Source
Drone can self-locate without GPS	DR: 3.4% drift over 800m/5min	DR feasibility test
Camera provides accurate target bearing	Bbox centroid → bearing: directionally correct	YOLO/SAMURAI tracking test
Target trajectory estimatable from camera	Smoothed bbox path matches real movement	DR cross-plot validation
Single drone cannot determine range	Bbox range ±30-50% — triangulation needed	Bbox size analysis
SAMURAI maintains track through gaps	269/394 frames tracked across lock/lost cycles	sea_drone_data_v1.json
IMU provides reliable heading for DR	Compass + yaw stable; supports 800m DR	IMU data analysis
Jetson can run full pipeline in real-time	118ms inference + tracking + control < 750ms/frame	Processing timing data
Mesh radio supports inter-drone comm	IP-based RF bridge on shared frequency	Network topology

Appendix A: Minimum Viable Data Packet

For severely bandwidth-constrained links (e.g., acoustic underwater):

```
MINIMAL_BEARING_PACKET (24 bytes):
  - x_position:     float32 (4 bytes)    // meters east from GS
  - y_position:     float32 (4 bytes)    // meters north from GS
  - heading:        float16 (2 bytes)
  - target_bearing: float16 (2 bytes)
  - bearing_rate:   float16 (2 bytes)
  - confidence:     uint8   (1 byte)
  - bbox_cx:        uint16  (2 bytes)
  - bbox_area:      uint16  (2 bytes)
  - lost_count:     uint8   (1 byte)
  - timestamp_ms:   uint32  (4 bytes)
```

Appendix B: MAVLink Integration

```
JETSON → PIXHAWK (control, 10-20 Hz):
  MAV_CMD: RC_OVERRIDE
    ch1 = steering PWM (1000-2000, center 1500)
    ch3 = throttle PWM (1000-2000)

PIXHAWK → JETSON (telemetry, 1-10 Hz):
  ATTITUDE:     roll, pitch, yaw (for bearing correction)
  VFR_HUD:      heading, groundspeed (for dead reckoning)
  GPS_RAW_INT:  lat, lon, satellites (for DR calibration when available)
  SYS_STATUS:   battery voltage, remaining percent
```

Appendix C: V5 Simulation Mapping

Real System	Simulation Equivalent
YOLOv8/SAMURAI detection	FOV cone check (inFOV function)
Bearing from bbox centroid	droneHeading + FOV geometry
Dead reckoning	posAt() position computation
BEARING_REPORT message	commAB/commBA flags + instruction log
D2D protocol	Cyan beam, direct packets
D2G protocol	Amber dashed beams via GS model
Ground Station	GS icon + CommDashboard + topology
Triangulation result	PathOptPanel (angular sep, GDOP)
TARGET_LOST event	Scenario 2, A_LOST phase
Pincer recovery	Coordinated search in A_LOST phase
Proportional navigation	B's acceleration curve toward intercept